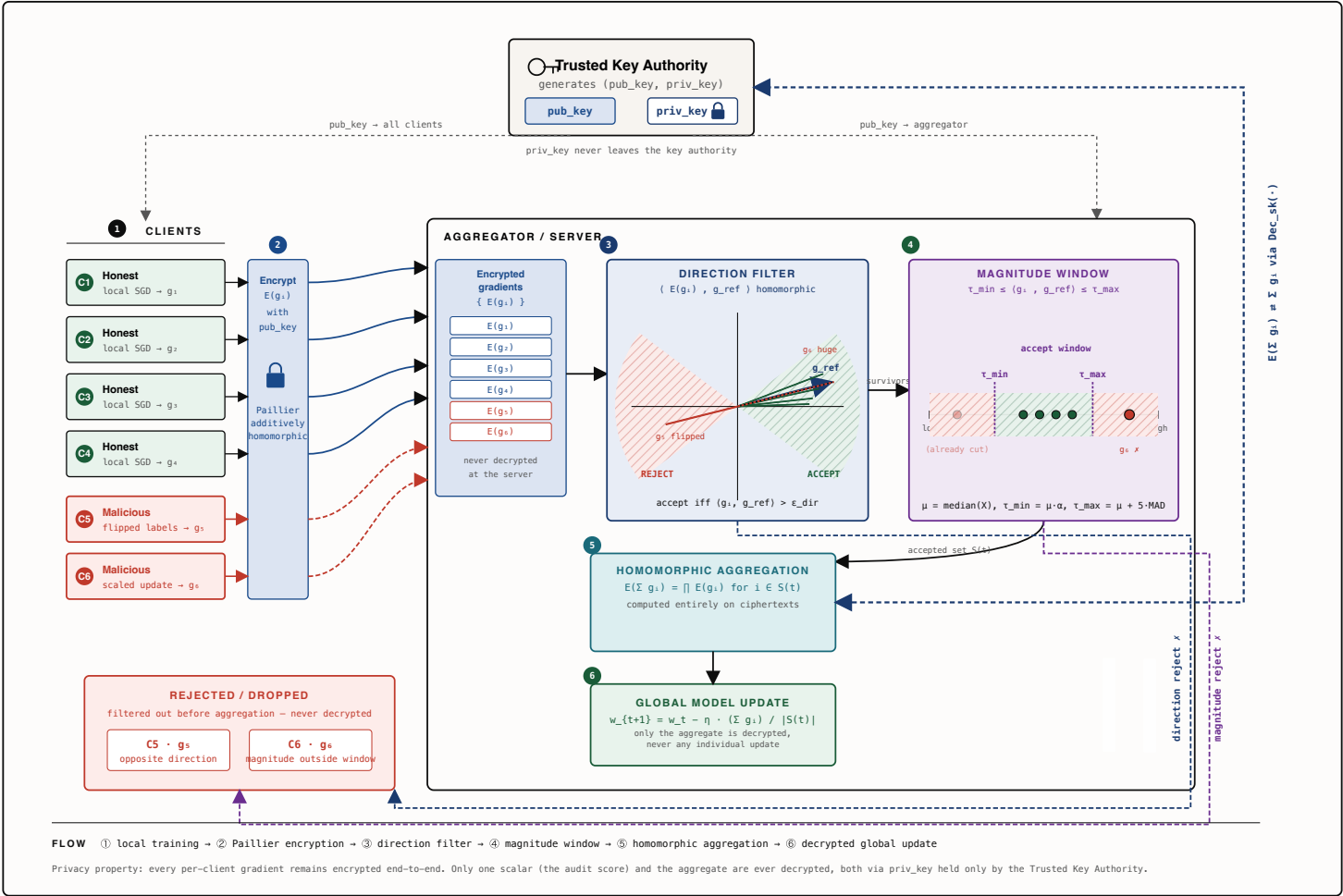


FIP-PPFL — Functional Inner Product · Privacy-Preserving Federated Learning

A trusted key authority publishes a Paillier public key. Clients train locally and submit encrypted gradients. The aggregator audits each encrypted update against a benign reference using two geometric criteria — first a directional check, then a magnitude window — and homomorphically aggregates only the survivors. The single private key never leaves the key authority.



① TRUSTED KEY AUTHORITY

Generates one Paillier keypair. Distributes the public key to all clients and the aggregator; keeps the private key sealed and uses it only for decryption of the audit score and the aggregate.

$pub_key \rightarrow$ broadcast
 $priv_key \rightarrow$ never leaves

③ DIRECTION FILTER

Server computes the homomorphic inner product $\langle g_i, g_{ref} \rangle$ against a benign reference built on public data. Updates pointing in the wrong half-space are removed before any aggregation.

accept iff $\langle g_i, g_{ref} \rangle > \epsilon_{dir}$

④ MAGNITUDE WINDOW

Robust statistics on the alignment scores. Median μ and MAD define an adaptive window; any client whose score is too small or too large is dropped — symmetric protection against shrinking and scaling attacks.

$\tau_{min} \leq \langle g_i, g_{ref} \rangle \leq \tau_{max}$

⑤ SECURE AGGREGATION

The encrypted gradients of the surviving set $S(t)$ are summed homomorphically. Only the aggregate is sent to the key authority for decryption — individual gradients never appear in plaintext.

$E(\sum g_i) = \sum E(g_i), i \in S(t)$